

Attention-Enhanced CNNs for Explainable Brain Tumor Classification with Grad-CAM and ROAD

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Abstract—Catching a brain tumor early changes everything about how a patient gets treated, but the diagnosis itself isn't easy to get right. Deep learning models can classify these tumors from MRI scans with decent accuracy these days. The problem is nobody can see why a model picked one answer over another, it just spits out a label. That's the gap this paper tries to close. Three backbones are tested here, ResNet50, DenseNet121, and EfficientNetB5, each paired with four attention setups (none, SE, CBAM, and a hybrid SE-CBAM combo), so twelve combinations total. The strongest three get retrained at a higher resolution afterward. For the winning model, Grad-CAM pulls out a heatmap of where it's actually looking, and the ROAD score checks whether that's a real signal or just noise. DenseNet121 with SE attention came out on top at 95

Index Terms—brain tumor classification, attention mechanisms, explainable AI, Grad-CAM, convolutional neural networks

I. INTRODUCTION

Brain tumors are one of the more dangerous health problems a person can face, and catching them early changes how a patient gets treated. MRI scans are the usual way doctors check for tumors, but reading them by hand takes time and depends a lot on the experience of the person looking at the scan, and different tumor types can sometimes look quite similar on a scan, making the diagnosis harder even for an experienced radiologist. Deep learning models, especially CNNs, came in to help with this, but most of them work like a black box, giving a label like "glioma" with no way for a doctor to check what part of the scan actually drove that decision.

A few specific findings from recent work have shaped how this paper has been set up. Guluwadi showed that pairing ResNet50 with Grad-CAM makes the predictions easier to trust, but the study was carried out only for tumor versus no-tumor classification, and not for the four-class version of the problem. [1]. Afnaan et al. added SE attention into ResNet101 and got a solid jump in accuracy across four tumor classes, but only tried SE on its own, without checking it against CBAM or a hybrid setup [2]. Jiang found that SE blocks reliably improve ResNet performance across different medical imaging tasks, though the gain can get unstable on smaller datasets [3]. Sankari et al. showed that different tumor types depend on different levels of clarity in a scan, which is why a single attention design might not work same for every tumor class

[5]. Ullah et al. tested both SE and CBAM across five different CNN backbones and found that EfficientNetB5 paired with a hybrid SE+CBAM setup gave the strongest results overall [16].

For the explainability, Chaddad et al. compared several Grad-CAM variants using the ROAD score instead of just visually analyzing the heatmaps, and found that this method gives a much more reliable answer than just a visual one [8]. Noor et al. and Patrício et al., in their reviews of explainable AI in healthcare, point out that Grad-CAM is the most commonly used method for CNN-based image analysis, but the field still lacks a standard way to measure if the explanation is actually trustworthy or not [17], [18].

Based on the observations, there are two gaps that needed focus. First, instead of testing one backbone with one attention type, multiple backbones needed to be tested against multiple attention types side by side, the way Ullah et al. did, but specifically for four-class brain tumor classification rather than a larger comparison. Second, instead of relying solely at a Grad-CAM heatmap and calling it explainable, the ROAD score needed to be included the way Chaddad et al. used it, to actually check if the model's attention is meaningful or not. This paper focuses on exactly these two points: three CNN backbones, ResNet50, DenseNet121, and EfficientNetB5, are tested with four attention setups each, no attention, SE, CBAM, and a hybrid SE-CBAM combination, giving twelve combinations in total. The strongest combinations are then re-trained at a higher resolution, and the best one is checked using both, Grad-CAM and ROAD score together, so that the explanation is supported by a number, not just through visual assessment alone.

II. PROPOSED METHODOLOGY

A. Overview

There are basically four things happening in this pipeline. Data gets prepped first, then the twelve backbone-attention combinations are built and trained. Then, the results get compared, and finally, whichever model wins goes through one more round just to check if its explanations can actually be trusted. Fig. 1 lays out this flow as a block diagram.

As Fig. 1 shows, an MRI image first gets resized and cleaned up, then passed through one of the three backbones. If the configuration includes an attention block, it sits right

before the final classification layer and re-weights the features coming out of the backbone. The model then outputs one of four labels. Once the strongest combination is identified, it goes through one more stage where Grad-CAM and the ROAD score check whether the model’s attention is actually landing on the tumor region or somewhere irrelevant.

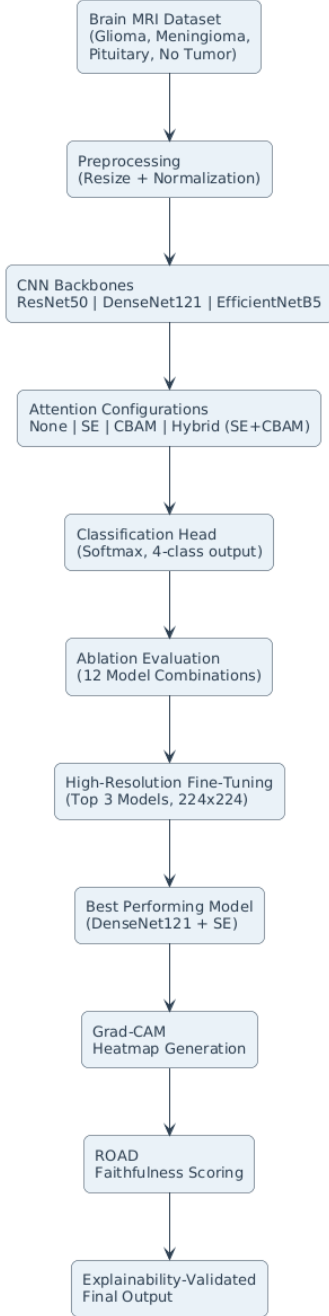


Fig. 1: Block diagram of the proposed attention-based classification and explainability framework.

B. Backbone Architectures

Three CNN backbones are used to extract features out of the MRI images. ResNet50 uses skip connections which

are shortcuts that let information skip over a few layers, which stops the network from forgetting useful signals as it gets deeper. DenseNet121 takes this further, where each layer receives information from all previous layers inside a dense block. Through this the model does not have to recompute features which allows the network to remain efficient while still learning detailed patterns from images. EfficientNetB5 takes a different approach, where the network’s depth, width, and input resolution is increased together in a balanced way instead of adding more layers. All three use the ImageNet weights as their starting point and are then fine-tuned specifically on the brain MRI data, which is a common transfer learning approach [11].

C. Attention Mechanisms

On top of each backbone, four attention setups are tested.

No attention is simply the backbone itself, used as the baseline to compare the effects of different attention mechanisms.

Squeeze-and-Excitation (SE) looks at each feature channel coming out of a layer, squeezes it down to a few using global average pooling, and uses that to decide how much weight each channel should get. It lets the network turn up or excite the channels that actually carry useful tumor information and turn down or squeeze the ones that don’t.

Convolutional Block Attention Module (CBAM) does the same channel-weighting step as SE, but then adds a second step called spatial weighting that looks at “where” in the image to focus, producing a small map that highlights useful spatial regions rather than just useful channels.

Hybrid attention runs SE and CBAM one after the other in the same block, the idea being that channel weighting and spatial weighting together might catch more than either one alone. Ullah et al. tested exactly this kind of SE+CBAM combination across five CNN backbones and found that it gave the strongest results specifically for an EfficientNetB5 backbone [16], which is part of why this combination is included here as well, to check if the same advantage of hybrid attention shows up on a different dataset and backbone setup.

D. Dataset and Preprocessing

The dataset used in this work is split into four classes: glioma, meningioma, pituitary tumor, and no tumor. Every image goes through resizing and normalization before being used in a model. The test set is balanced with 400 images of each class. The first round of testing, covering all twelve backbone-attention combinations, uses a smaller 128×128 input size with a batch size of 16. This is mainly done to keep the training fast enough to compare that many configurations in a reasonable amount of time. Once the top three combinations are identified, they are retrained at a larger 224×224 size with a batch size of 32 since a bigger image gives the network more details to work on which usually increases the accuracy.

E. Implementation Details

All models are trained using the Adam optimizer with a starting learning rate of 1×10^{-4} . After the global average

pooling layer, a dropout of 0.4 is applied, followed by a dense layer with 256 units, then a second dropout of 0.3 before the final classification layer, which helps reduce overfitting given the relatively small dataset size. Rather than training every layer from the start, the backbone is initially kept frozen while only the attention block and classification head learn, after which roughly half of the network, starting from the earliest layers, is unfrozen for fine-tuning, while the deeper layers closer to the output continue training throughout. Training runs for up to 30 epochs, but rarely needs all of them, since early stopping is used to halt training if validation accuracy does not improve for 7 consecutive epochs, restoring the best-performing weights once it stops. The learning rate is also automatically reduced by half whenever validation loss plateaus for 3 epochs, helping the model settle into a better minimum during the later stages of training. During training, data augmentation, random rotation, width and height shifts, shear, zoom, and horizontal flipping, is applied to the training images to make the model less sensitive to minor variations in scan positioning, and 20% of the training data is set aside as a validation split to monitor performance during training.

F. Evaluation Metrics

A few standard metrics are used to judge how well each model performs. *Accuracy* is just the percentage of test images the model got right overall. *Precision*, for any one class, checks out of everything the model labeled as that class, how many were actually correct. *Recall* checks the opposite direction, out of everything that truly belonged to that class, how many the model actually caught. *F1-score* balances precision and recall into one number, which is useful when a model is doing noticeably better on one class than another.

Additionally, two more things are used to check if the best model can actually be trusted and is not just accurate. Grad-CAM works by looking at how strongly the final prediction depends on each region of the feature map, then turns that into a heatmap showing which part of the scan mattered most. Just looking at a heatmap and saying "yeah, that looks about right" isn't strong enough, which is exactly the gap Chaddad et al. pointed out when they compared several Grad-CAM-style methods using an actual score instead of relying on visual judgment [8]. So, the ROAD score is used here too. It works by removing the pixels the heatmap says are most important and checking how much the model's confidence drops; if it drops a lot, the heatmap was pointing at something real, and if it barely changes, the heatmap wasn't actually telling us much.

III. RESULTS AND DISCUSSION

A. Dataset Description

The dataset contains four diagnostic categories, glioma, meningioma, pituitary tumor, and no tumor. They were collected from publicly available brain MRI collections where each image is a single MRI slice, either an axial or sagittal view. The dataset is already split into a training portion and a test portion. The test portion is kept balanced, 400 images

for each of the four classes, 1600 in total, so that none of the classes get an unfair advantage when accuracy is calculated. Fig. 2 shows one example image from each of the four classes, just to give a sense of how visually close some of these categories can look, even to a trained eye.

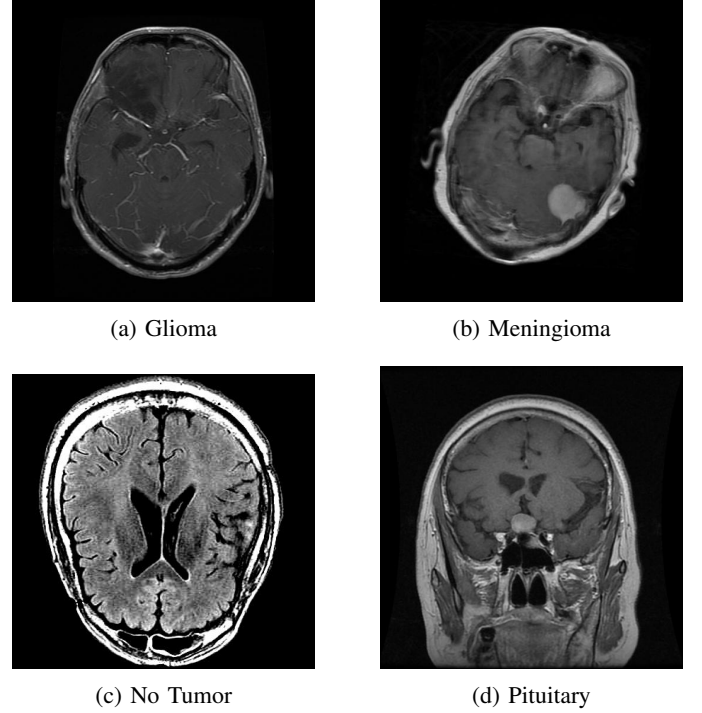


Fig. 2: Sample MRI images from each of the four diagnostic classes.

Before training, we resize and normalize all the images first. We start with 128×128 because training all twelve backbone attention combinations at a higher resolution would take longer time. Once we figure out which three combinations perform the best, we retrain them at 224×224 to see if they can squeeze out some extra accuracy, just like we discussed in the Methodology section.

B. ResNet50-Based Models

We test four versions of ResNet50: the plain model, SE attention, CBAM, and hybrid attention. As shown in Table I, the plain ResNet50 achieves an accuracy of 82.38%, and adding SE attention improves it to 85.44%. On the other hand, CBAM performs worse than the baseline with 78.25%, and the hybrid model ends up with a similar accuracy of 82.44%. From these results, it is clear that SE attention works well with ResNet50, whereas CBAM does not seem to capture any useful features from the brain MRI images. The hybrid model also does not provide any noticeable improvement. The overall training process followed for all four variants is shown in Algorithm 1.

C. DenseNet121-Based Models

DenseNet121 turns out to be the best-performing backbone among the three. Even without any attention module, it

Algorithm 1 ResNet50-Based Model Training Procedure

- 1: Load ImageNet-pretrained ResNet50 weights
- 2: Remove the original classification head
- 3: **if** attention configuration is SE, CBAM, or Hybrid **then**
- 4: Insert the corresponding attention block(s) after the final convolutional stage
- 5: **end if**
- 6: Add global average pooling and a dense softmax layer with 4 output units
- 7: Freeze the early convolutional layers; keep the final stages trainable
- 8: **for** each training epoch **do**
- 9: Forward-propagate a batch through the backbone and attention block
- 10: Compute the categorical cross-entropy loss against the true labels
- 11: Backpropagate and update the trainable weights
- 12: **end for**
- 13: Evaluate on the held-out test set

achieves an accuracy of 93.56%. Adding SE attention further improves it to 93.94%, which is the highest accuracy among all twelve model combinations. CBAM and hybrid attention also perform well, but their accuracies are slightly lower at 93.31% and 93.25%, respectively. Similar to what we observed with ResNet50, SE attention is the only module that consistently improves the model’s performance, while CBAM and hybrid attention do not provide any significant gain. The training process remains the same as Algorithm 1, with DenseNet121 used as the backbone instead of ResNet50.

D. EfficientNetB5-Based Models

EfficientNetB5 lands in between the other two backbones. The plain version gets 90.75%, SE attention takes it to 92.31%, and CBAM and hybrid come in at 91.25% and 91.81% respectively. Once again, SE gives the most improvement, which is the same trend followed by the other 2 backbones too.

TABLE I: Ablation Study Results at 128×128 Resolution

ID	Backbone	Attention	Accuracy
A1	EfficientNetB5	None	90.75%
A2	EfficientNetB5	SE	92.31%
A3	EfficientNetB5	CBAM	91.25%
A4	EfficientNetB5	Hybrid	91.81%
B1	ResNet50	None	82.38%
B2	ResNet50	SE	85.44%
B3	ResNet50	CBAM	78.25%
B4	ResNet50	Hybrid	82.44%
C1	DenseNet121	None	93.56%
C2	DenseNet121	SE	93.94%
C3	DenseNet121	CBAM	93.31%
C4	DenseNet121	Hybrid	93.25%

E. High-Resolution Fine-Tuning of Top Candidates

Based on these results, the three best combinations, DenseNet121+SE, plain DenseNet121, and Efficient-

NetB5+SE, are picked for another round of training, this time at 224×224 . Table II shows DenseNet121+SE reaching 95.00% accuracy, the strongest result among the three. Plain DenseNet121 reaches 94.81%, and EfficientNetB5+SE reaches 93.50%. The same ranking from the smaller-resolution test holds up here too, so DenseNet121 with SE attention is confirmed as the best combination overall.

TABLE II: Fine-Tuning Results at 224×224 Resolution

ID	Backbone	Attention	Accuracy
C2	DenseNet121	SE	95.00%
C1	DenseNet121	None	94.81%
A2	EfficientNetB5	SE	93.50%

Fig. 3 shows how training and validation accuracy change across epochs for the best model during this fine-tuning stage. The two curves stay quite close to each other through most of training, which is generally a good sign, it suggests the model is picking up on real patterns in the data rather than just memorizing the training images.

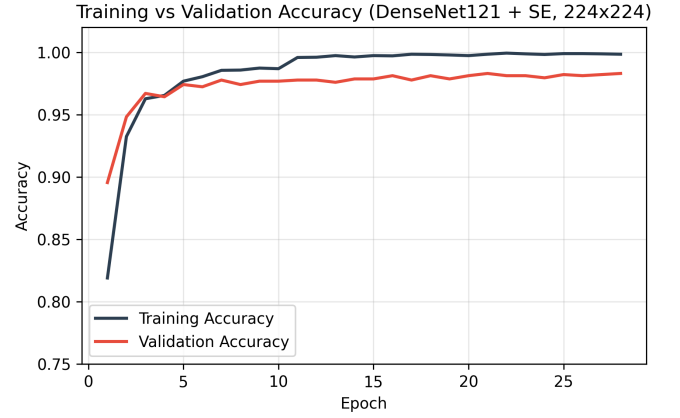
Fig. 3: Training and validation accuracy of DenseNet121+SE across training epochs at 224×224 resolution.

Fig. 4 compares validation accuracy across all three fine-tuned models on the same plot. DenseNet121+SE and the plain DenseNet121 baseline move closely together and climb faster from the beginning, while EfficientNetB5+SE starts much lower and takes longer to catch up, eventually forming a plateau under the other two.

F. Classification Performance of the Best Model

Table III shows the precision, recall, and F1-score of the DenseNet121 and SE model for all four classes on the 1600 image test set. The model achieves perfect precision of 1 for the glioma class, but its recall is slightly lower at 0.84, which means a few glioma images are classified as other tumor types. The no-tumor and pituitary classes both achieve a perfect recall of 1, showing that the model identifies these classes very well. Overall, the model achieves an average F1-score of 0.95 across all four classes, indicating strong and balanced performance.

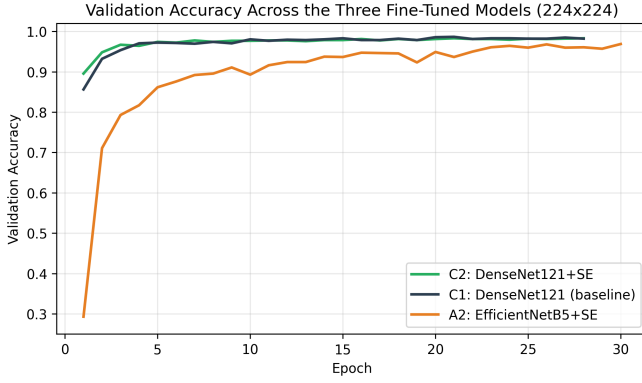


Fig. 4: Validation accuracy across training epochs for all three fine-tuned models at 224×224 resolution.

TABLE III: Classification Performance of DenseNet121+SE (224×224)

Class	Precision	Recall	F1	Support
Glioma	1.00	0.84	0.91	400
Meningioma	0.92	0.96	0.94	400
No Tumor	0.93	1.00	0.96	400
Pituitary	0.96	1.00	0.98	400
Macro Avg	0.95	0.95	0.95	1600

G. Explainability Validation Using Grad-CAM and ROAD

We also use Grad-CAM to understand how the best-performing model makes its predictions. For every test image, Grad-CAM generates a heatmap that highlights the parts of the MRI scan the model pays the most attention to. Fig. 6 shows one example from each of the four classes along with the original MRI image and its corresponding heatmap. To evaluate whether these highlighted regions are actually useful, we calculate the ROAD score for each class using the same method followed by Chaddad et al. [8]. The results are listed

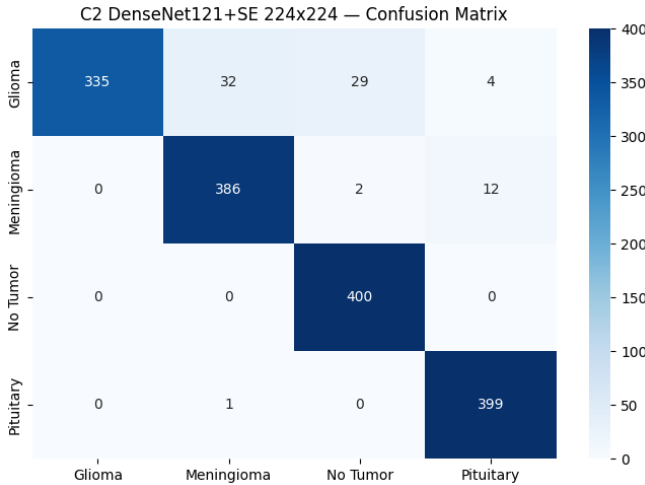


Fig. 5: Confusion matrix of the DenseNet121+SE model on the test set.

in Table IV. The model achieves an average ROAD score of 0.4205 across all classes, indicating that the highlighted regions contribute meaningfully to its predictions. Among the four classes, pituitary records the highest score of 0.6231, followed by meningioma with 0.5822, which is expected since these tumors usually appear as clearly defined regions in MRI scans. Glioma obtains a lower score of 0.320 because these tumors often spread over a larger area instead of forming a single well-defined region, making them harder for Grad-CAM to highlight effectively, as also reported by Sankari et al. [5]. The no-tumor class has the lowest ROAD score of 0.1567, which is reasonable because there is no tumor region for the model to focus on.

TABLE IV: Per-Class ROAD Faithfulness Scores

Class	ROAD Score
Pituitary	0.6231
Meningioma	0.5822
Glioma	0.3200
No Tumor	0.1567
Mean	0.4205

H. Discussion

Looking at the results across all three backbones, one pattern is very clear. SE attention consistently improves the performance, CBAM gives mixed results, and the hybrid attention model never performs better than SE alone. This is different from what Ullah et al. [16] reported, where combining SE and CBAM gave the best performance with EfficientNetB5. A possible reason for this difference is that brain MRI images contain more useful information in the feature channels than in distinct spatial regions. Because of this, the additional spatial attention used by CBAM does not seem to provide much benefit for our dataset, even though it worked well in their study.

Another clear observation is that DenseNet121 performs better than both ResNet50 and EfficientNetB0 across all attention modules. This is consistent with the findings of Erukude et al. [13], who also reported better performance with DenseNet121. The dense connections in DenseNet121 likely help preserve important tumor features throughout the network, which may explain its better performance. The ROAD scores also support our results by showing that the model is not only achieving high accuracy but is also focusing on meaningful regions of the MRI images while making predictions.

Some of the previous studies also point towards possible future work. For example, Islam et al. [14] showed that Xception outperformed ResNet50 on a similar four-class brain MRI dataset without using any attention mechanism. Since Xception was not included in our experiments, it would be interesting to evaluate it with different attention modules in future work and compare its performance with DenseNet121.

Grad-CAM Explainability — DenseNet121 + SE Attention
(Best Model: 95.00% Test Accuracy)

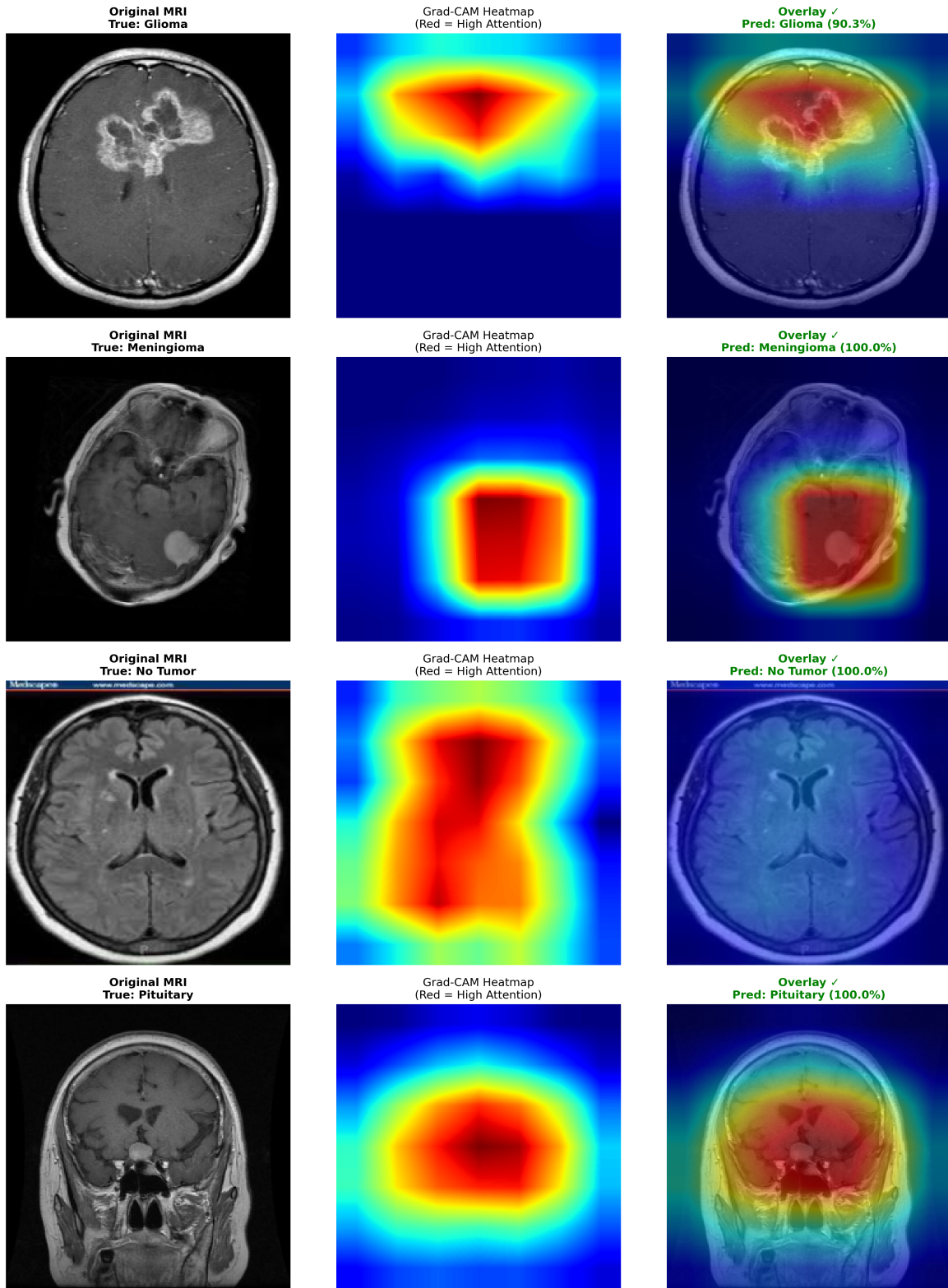


Fig. 6: Grad-CAM heatmaps and prediction overlays for one sample image from each of the four diagnostic classes, showing the original MRI scan, the corresponding heatmap, and the resulting overlay with predicted label and confidence.

I. Comparison with Existing Approaches

Table V shows how our work compares to the accuracy figures reported in some of the literature reviewed earlier. These numbers are not perfectly comparable, since the studies differ in dataset, number of classes, and sometimes whether the task is binary or four-class, still the comparison is useful context. Guluwadi reports a higher number, but only for binary tumor versus no-tumor classification, which is by default an easier task [1]. Afnaan et al., Aiya et al., Sankari et al., and Islam et al. all report results in the high 90s for four-class classification, similar to or slightly above what is achieved here [2], [4], [5], [14]. Erukude et al.'s plain DenseNet121, without any attention mechanism, sits below this work's result [13], supporting the case that adding SE attention does improve the performance of the model in this setup. The other studies, however, focus more on the classification performance more than the explainability evaluation whereas our work does focus on it using the ROAD score, which is a meaningful point of difference rather than accuracy alone.

TABLE V: Comparison with Accuracy Reported in Reviewed Literature

Study	Task	Accuracy
This work (DenseNet121+SE)	4-class	95.00%
Guluwadi [1]	Binary	98.52%
Afnaan et al. [2]	4-class	97.27–99.65%
Aiya et al. [4]	4-class	~99%
Sankari et al. [5]	4-class	98.7%
Erukude et al. [13]	Binary	94.3%
Islam et al. [14]	4-class	98.71%

J. Computational Efficiency

Beyond accuracy, the time needed to train each backbone is also worth noting. Both DenseNet121 variants, with and without SE attention, trained at roughly 80 seconds per epoch on the available GPU. EfficientNetB5 took noticeably longer, generally between 95 and 105 seconds per epoch, which is expected as it has a larger network with more parameters to update. Given that it takes more time, EfficientNetB5+SE still ended up with lower accuracy (93.50%) than DenseNet121+SE (95.00%). DenseNet121 turns out to be the better choice on both factors here, not just more accurate but also faster to train, which matters if this kind of model were ever retrained regularly on an updates or a different dataset.

K. Limitations

This work is tested on one curated MRI dataset, so how well it would perform on scans from a different hospital or a different scanner is not yet known. The ROAD check is also only done on the single best model(DenseNet121 + SE). Comparing faithfulness scores across all twelve combinations was not done due to compute limits. Both of these are reasonable directions to expand on later.

IV. CONCLUSION AND FUTURE SCOPE OF WORK

This work compares three CNN backbones, ResNet50, DenseNet121, and EfficientNetB5, across four attention setups, no attention, SE, CBAM, and a hybrid SE-CBAM combination, giving twelve combinations in total for four-class brain tumor classification covering glioma, meningioma, pituitary tumor, and no tumor. Across all three backbones, SE attention is the one setup that consistently helps, while CBAM is unreliable and sometimes drops accuracy below the plain baseline, and the hybrid combination never performed better than SE on its own. DenseNet121 paired with SE attention comes out as the strongest combination overall, reaching 95.00% accuracy after retraining at 224×224 resolution, with a macro-averaged F1-score of 0.95 across all four classes. Beyond the accuracy numbers, checking this model with Grad-CAM and the ROAD score shows that it is not just performing well but also the regions it focuses on for pituitary and meningioma cases align with what would actually be expected on a scan. Even where the score comes out lower, for glioma and no-tumor, there is a reasonable explanation behind it tied to how those classes naturally appear. DenseNet121 also trains noticeably faster than EfficientNetB5 while still coming out more accurate, which makes it the better choice here on both factors and not just one.

Future work can extend this in a few directions. Testing the same setup on MRI scans from a different hospital or scanner would help confirm whether the results hold up outside this one dataset. Newer self-explainable architectures where interpretability is built into the model itself rather than added after prediction, could also be explored in future work [19]. Running the ROAD check across all twelve combinations, not just the best one, would also clarify whether explainability and accuracy improve together or independently. Also, since attention-based designs have shown strong results on other imaging types such as skin lesions and CT scans [6], [9], [12], checking whether this exact framework performs well on other imaging tasks is a reasonable next step.

Table VI gives a condensed view of the literature discussed throughout this paper, listing the method and main finding or gap for each.

TABLE VI: Summary of Reviewed Literature

Ref	Author(s)	Method / Focus	Key Finding or Gap
1	Guluwadi	ResNet50 + Grad-CAM, binary tumor classification	98.52% accuracy; binary only, no attention module tested
2	Afnaan et al.	SE-ResNet101 + Grad-CAM/LIME, 4-class	Strong accuracy; only SE tested, no comparison with CBAM or hybrid
3	Jiang	SE blocks in ResNet-18 across medical imaging tasks	SE improves results but can pick up noise on small datasets
4	Aiyya et al.	VGG16 + custom channel attention, radiologist-validated	Near 99% accuracy; no ablation across attention types
5	Sankari et al.	Multi-scale Vision Transformer, brain MRI	98.7% accuracy; different tumor types need different detail scales
6	Himel et al.	SAM + ViT, skin cancer segmentation	96.15% accuracy; binary classification only
7	Krishnan et al.	Rotation-invariant ViT, brain MRI	Removes need for rotation augmentation; binary, visual-only explainability
8	Chaddad et al.	GradCAM++/LayerCAM/XGradCAM compared using ROAD	First to quantify explainability; tested only on plain ResNet50
9	Zhu and Yang	Dual-attention ResNet, abdominal CT quality	Works across ResNet depths; single CT dataset only
10	Zhang et al.	Review of attention mechanisms in segmentation	Highlights lack of standard evaluation across the field
11	Thakur et al.	Broad review of DL in medical imaging	Notes generalization and explainability as open problems
12	Al-Waisy et al.	Skin-DeepNet, multi-stage skin cancer pipeline	Near-perfect accuracy; high architectural complexity
13	Erukude et al.	ResNet50 vs. DenseNet121, Grad-CAM	DenseNet121 wins (94.3%); no attention tested on either
14	Islam et al.	Xception vs. ResNet50, 4-class brain MRI	Xception best (98.71%); no attention-based enhancement
15	Ahmad et al.	Transfer learning benchmarks, chest X-ray	InceptionV3 best but computationally heavy
16	Ullah et al.	SE + CBAM across 5 backbones, 2 datasets	Hybrid SE+CBAM best for EfficientNetB5
17	Noor et al.	Review of 89 XAI studies in healthcare	Grad-CAM dominant; no standard quantitative check
18	Patrício et al.	Survey of Grad-CAM, LIME, SHAP, LRP	Grad-CAM popular but can highlight irrelevant regions
19	Hou et al.	Survey of self-explainable architectures	Promising direction; not yet tested on medical data

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